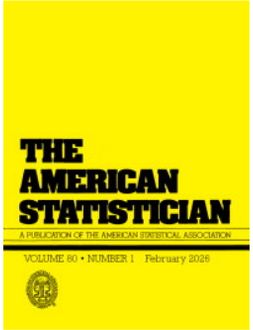

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Abstract

Laboratories use group testing to test high volumes of clinical specimens for pathogens, such as SARS-CoV-2, West Nile, and Chlamydia trachomatis. The process works by testing multiple specimens together as an amalgamation, rather than testing each specimen separately, to reduce the number of tests needed. There are many different algorithmic ways to apply group testing. The role of a statistician is to recommend an algorithm that will perform "best" relative to the information available, such as disease prevalence. Algorithms are most often compared by their expected number of tests needed for an application, where a lower value is preferred. Unfortunately, this measure alone does not account for some algorithms having a lower expected number of tests at the expense of being much more complex to implement. For this reason, we propose a new measure that we refer to as the *complexity*. In our paper, we present its definition and derive its expression for several common algorithms. We show that some algorithms may be too complex for implementation, while others should become more widely used. Our proposed measure is illustrated with a SARS-CoV-2 testing implementation. R functions and a Shiny app are provided to perform calculations.

Keywords: Array testing; Case identification; COVID-19; Disease testing; Efficiency; Pooled testing

1 Introduction

Timely and efficient testing is essential to identify individuals who have an infectious disease. This was abundantly clear early on during the COVID-19 pandemic when testing was delayed or even unavailable, likely resulting in increased disease spread. Fortunately, widespread use of group testing (also known as pooled testing) eventually helped manage the overwhelming number of specimens received by laboratories. Its use was covered extensively by the news media (e.g., Bui et al. 2020, Gossner and Gollier 2020, Seucharan 2020), while also becoming pervasive with the issuance of Emergency Use Authorizations (EUAs; U.S. Food & Drug Administration 2025). Group testing research was abundant during the pandemic with more than

1 GB of papers published during its first two years. This research showed how to implement group testing as well as its effectiveness (e.g., Denny et al. 2020, Yelin et al. 2020, Pollock et al. 2021).

Group testing is implemented through a series of sequential stages that can be formulated into an algorithm. The simplest and most widely-used algorithm is referred to as Dorfman testing, which is named after Robert Dorfman and his seminal paper on group testing (Dorfman 1943). Abdalhamid et al. (2020) was likely the first community-based implementation of this algorithm during the COVID-19 pandemic. In their first stage of the algorithm, they combined portions of specimens from five individuals to form a group and tested the group using a reverse transcription polymerase chain reaction (RT-PCR) assay developed by the Centers for Disease Control and Prevention (CDC). If a group tested negative for SARS-CoV-2, all individuals within the group were declared negative for the presence of the virus. Instead, if a group tested positive for SARS-CoV-2, each individual was retested separately in a second stage using new portions of their original specimens. This resulted in separate positive or negative outcomes for each individual. Dorfman testing was applied around the world, where the same or different group sizes were used (e.g., 4 for Mobini et al. 2023, 10 for Smith et al. 2023, and 20 for Alvargonzalez et al. 2024). Other group testing implementations sometimes used different algorithms, which we discuss further in Section 2. Some of these algorithms involved just two stages, like Dorfman's, but some algorithms involved many more.

An important role of a statistician is to recommend a group testing algorithm to a laboratory that will perform "best" in some way. This recommendation is based on factors like disease prevalence, assay accuracy, and laboratory constraints. The focus for "best" typically centers around the expected number of tests needed per individual, often referred to as the *efficiency* (e.g., Kim et al. 2007, Lendle et al. 2012, Comess et al. 2022). For example, an efficiency of 0.25 represents that only a quarter of the number of tests on average would be needed for a group testing algorithm when compared to testing each specimen separately. This also represents a 75% reduction in the expected number of tests per individual and a 300% increase in expected testing capacity (assuming a continuous supply of specimens) when compared to testing specimens separately (Bilder et al. 2020). The statistician would attempt to find the algorithm that has the best efficiency (i.e., the smallest expected number of tests per individual). Thus, given two group testing algorithms with their efficiency values, one would prefer the algorithm with the lowest value as long as the overall diagnostic accuracy for the algorithm is sufficiently large.

A potential problem with this approach is some algorithms may not be realistic for laboratory implementation. In particular, an algorithm could have good efficiency but be difficult to put into practice. Fenichel et al. (2021) highlight this problem by saying,

While statisticians, operations researchers, and others with expertise in sampling [algorithm] design have important value to add, end users [laboratories] will often identify real-world constraints that must be accounted for during design; statisticians and operations researchers often appear unaware of these constraints. This suggests that there are still substantial opportunities for these communities to collaborate and enhance testing capacity, as it may be surprising to those designing pooling strategies to learn that laboratories view pooling as more time consuming, delaying test reporting.

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Our long-time collaborations with public health officials and lab technicians have produced conversations similar to the concerns conveyed by Fenichel et al. (2021). This has motivated us to look for statistical ways that can explain how complex a group testing algorithm is to implement. The focus in our paper is on computing the expected number of stages for a particular individual to be declared positive or negative for a pathogen. We refer to this quantity as the *complexity* for an algorithm. The reason for the name is that each additional stage increases the demands on a laboratory. Our purpose is to derive expressions for the complexity corresponding to common group testing algorithms. We will compute the complexity for a number of scenarios to determine which algorithms are practical for everyday use and which algorithms may be too complex for real-world implementation.

A key aspect of the complexity is that it can be converted to an expected amount of time. This is because each stage has a known amount of time it takes to complete, such as four hours for the CDC's assay to detect SARS-CoV-2 (Anderson 2020). Also, thinking of complexity relative to time is important not only for laboratories, but for the individuals being tested because it can provide an amount of time needed to obtain test results.

Throughout our paper, we focus on COVID-19 examples. However, this new measure can be used as well in the wide range of applications where group testing is implemented. For instance, group testing is used in other settings for human infection detection, including recent applications for blood donations (American Red Cross 2025), congenital infections (Merav et al. 2024), cancer (Wiesel et al. 2023), and malaria (Chang et al. 2021). Outside of human disease settings, group testing is used in other applications, including animal infection detection (Valas et al. 2023), human exposure to harmful chemical compounds (Heffernan et al. 2015), food safety surveillance (Nobrega et al. 2023), and computer network assessments (Ma et al. 2021).

The outline of our paper is as follows. Section 2 examines the formal definition of complexity and how to interpret it. We derive the complexity for five separate group testing algorithms in this section. These derivations purposely take advantage of previous research on efficiency to relate these measures and to simplify expressions. Section 3 examines the computation of the complexity for different disease prevalences and assay accuracies, while taking into account practical limitations that may be present. We make comparisons across these algorithms to understand which are perhaps too complex for implementation. Section 4 applies the complexity to a SARS-CoV-2 testing situation occurring for LabCorp. This company obtained an EUA for a group testing algorithm that had an efficiency worse than that for Dorfman testing. Section 5 summarizes our findings and discusses other non-measurable issues that may make group testing implementation difficult.

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2 Complexity

Consider an infectious disease testing scenario with an overall infection prevalence of p and an essentially limitless number of specimens available at a laboratory for testing. Suppose a set of I specimens are chosen from these specimens and are tested through a group testing algorithm. We assume independence among these specimens. Define T as the number of tests needed to determine which corresponding individuals are positive or negative for the pathogen of interest. The efficiency of the algorithm is defined as $E(T)/I$; i.e., the expected number of tests per

individual. The reason for dividing $E(T)$ by I is to allow for comparisons among algorithms that may use different values for I. This division then puts the comparison on a per-individual basis. As remarked in Section 1, the "best" algorithm is most often thought of as the one with the smallest $E(T)/I$.

Define C_i as the number of stages for the ith individual ($i = 1, \ldots, I$) to be declared positive or negative. The complexity of the algorithm for the ith individual is defined as $E(C_i)$. This measure is not divided by I because it is already on an individual basis. We also can relate $E(C_i)$ to the amount of time it takes to complete an algorithm. When the same assay is applied at each stage, denote a constant τ as the amount of time it takes to perform a test. The expected amount of time becomes $\tau E(C_i)$. When a different assay is used in some stages, the expected amount of time takes the same general form with modifications that we discuss in Web Appendix A. We will generally focus on complexity rather than time in our paper though to allow for general discussions relative to any assay.

Similar to the efficiency, algorithms with lower values for complexity in comparison to others are preferred. However, one should not choose an algorithm based on complexity alone without taking into account the efficiency as well. An extreme example to demonstrate this point is to test each individual separately, say, "individual testing". In that case, $E(C_i) = 1$, which is smaller than what can be obtained from a group testing algorithm. Individual testing's efficiency though likely will be much worse in the standard situations where group testing is used (p is small; see Hughes-Oliver 2006). We will examine this point again in Sections 3 and 4.

In the subsections that follow, we derive $E(C_i)$ for algorithms that fit into two classifications: hierarchical and non-hierarchical. Hierarchical algorithms obtain their name from allowing a specimen to be tested in a group only once per stage (e.g., Dorfman testing). Non-hierarchical algorithms allow specimens to be tested more than once per stage. Brief descriptions of specific algorithms are provided next with the derivations.

2.1 Hierarchical

The simplest hierarchical algorithm is the previously described Dorfman testing. This algorithm can also be referred to as two-stage hierarchical testing because its maximum number of stages is two. In general, hierarchical testing can include more than two stages. For example, Lohse et al. (2020) employed a three-stage version when testing specimens for SARS-CoV-2. In their first stage, $I = 30$ individuals were tested in a group. If this group tested positive, three new groups of size 10 were formed in stage two so that an individual was represented in only one of them. If a new group tested positive in the second stage, each member of that group was retested separately in a third stage to determine positive/negative outcomes. If at any stage a group tested negative, its members were declared negative. For further introductory information on hierarchical testing, see Kim et al. (2007) and Bilder (2019).

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Define $G_{d,i}$ as the binary testing outcome (1 = positive, 0 = negative) for the group that includes the specimen from individual *i* at stage *d*, where $d = 1, \ldots, D$. Also, define I_d as the group size for individual *i* at stage *d*, where $I \equiv I_1$. We first focus on Dorfman testing by stating the efficiency as given by Johnson et al. (1991) and Kim et al. (2007):

$$E(T)/I = \{1 + I \times P(G_{1,i} = 1)\}/I. \quad (1)$$

The expression within braces represents that one test is performed always for the stage one group and *I* further tests are performed if the group tests positive. To state the expression for $P(G_{1,i} = 1)$, define $S_{e,d}$ and $S_{p,d}$ as the sensitivity and specificity of the assay, respectively, at stage *d*. The probability is $P(G_{1,i} = 1) = (1 - S_{p,1})(1-p)^I + S_{e,1}\{1-(1-p)^I\}$.

Our proposed complexity measure for Dorfman testing is

$$E(C_i) = 1 + 1 \times P(G_{1,i} = 1). \quad (2)$$

This expression is similar in form to the efficiency. In particular, we see that the *I* within the braces of Equation 1 is replaced by a 1 because one more stage is needed if the stage one group is positive. Building on this intuitive reasoning, we state the complexity for two or higher stages as

$$\begin{aligned} E(C_i) \quad &= 1 + P(G_{1,i} = 1) + P(G_{1,i} = 1, G_{2,i} = 1) + \cdots \\ &\quad + P(G_{1,i} = 1, G_{2,i} = 1, \ldots, G_{D-1,i} = 1). \end{aligned} \quad (3)$$

Expressions for the probabilities in Equation 3 are available from Johnson et al. (1991) and Kim et al. (2007), because they were needed in their derivations of $E(T)$. It is important to note that their expressions are for the case of $S_{e,1} = \cdots = S_{e,D}$ and $S_{p,1} = \cdots = S_{p,D}$. While equal sensitivities and specificities across stages can occur in practice, there are some applications where it may not because of the dilution effect that sometimes occurs with group testing (Bilder et al. 2021*a*). We state here the expression for $E(C_i)$ that allow sensitivities and specificities to change across stages for three-stage hierarchical testing:

$$\begin{aligned} E(C_i) \quad &= 1 + P(G_{1,i} = 1) + P(G_{1,i} = 1, G_{2,i} = 1) \\ &= 1 + (1 - S_{p,1})(1-p)^{I_1} + S_{e,1}\left\{1 - (1-p)^{I_1}\right\} + (1 - S_{p,1})(1 - S_{p,2})(1-p)^{I_1} \quad (4) \\ &\quad + S_{e,1}(1 - S_{p,2})\left\{(1-p)^{I_2} - (1-p)^{I_1}\right\} + S_{e,1}S_{e,2}\left\{1 - (1-p)^{I_2}\right\}. \end{aligned}$$

The expression for $P(G_{1,i} = 1, G_{2,i} = 1)$ used in Equation 4 was first given by Hitt (2020) in their derivation of $E(T)$. Web Appendix A provides the generalization of Equation 4 to four-stage and higher hierarchical testing.

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2.2 Non-hierarchical

Non-hierarchical testing algorithms allow for an individual to be tested in more than one group during a stage. The most widely used of these testing algorithms is array testing (also known as matrix pooling), so we will focus on this algorithm here. The algorithm begins by arranging specimens into a grid. Groups are formed for each row of the grid and for each column of the grid, so that each specimen is within one row and one column group. Specimens at the intersection of positive testing row and column groups are retested in stage two to determine positive or negative outcomes. For those specimens not at an intersection, they are declared negative with one exception. If rows (columns) test positive without at least one column (row) testing positive, this indicates a testing error occurred, so each member of positive testing rows (columns) are retested separately. LabCorp received an EUA during the COVID-19 pandemic to implement array testing (LabCorp 2022), where we describe their procedure further in Web Appendix B. For further introductory information on array testing algorithms, see Kim et al. (2007) and Bilder (2019).

Suppose array testing uses a $J \times K$ grid of specimens. Note that the number of individuals represented in the array is $I = JK$. Define $\mathcal{R}_j$ as the binary testing outcome (1 = positive, 0 = negative) for the jth row group. Similarly, define $\mathcal{C}_k$ as the binary testing outcome for the kth column group. Kim et al. (2007) were the first to derive the efficiency which allowed for testing error in an array, but for $J = K$ (square array), $S_{e,1} = S_{e,2}$, and $S_{p,1} = S_{p,2}$. Hitt (2020) extended this to allow for $J \neq K$, $S_{e,1} \neq S_{e,2}$, and $S_{p,1} \neq S_{p,2}$. Based on these works, the efficiency for array testing is

$$E(T)/I = I^{-1}\left\{J + K + \sum_{j=1}^{J}\sum_{k=1}^{K} P(T_{jk} = 1)\right\}, \quad (5)$$

where T_{jk} is an indicator variable for whether the specimen in the jth row and kth column needs to be retested in a second stage. Formally, T_{jk} is written as

$$T_{jk} = \begin{cases} 1, & \text{if } \mathcal{R}_j = 1 \text{ and } \mathcal{C}_k = 1 \\ 1, & \text{if } \mathcal{R}_j = 1 \text{ and } \sum_{k'=1}^{K} \mathcal{C}_{k'} = 0 \\ 1, & \text{if } \sum_{j'=1}^{J} \mathcal{R}_{j'} = 0 \text{ and } \mathcal{C}_k = 1 \\ 0, & \text{otherwise.} \end{cases}$$

Because Kim et al. (2007) and Hitt (2020) derive the expression for $P(T_{jk} = 1)$, we do not reproduce it here but provide the expression in Web Appendix A.

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The complexity for array testing has a form similar to that for the efficiency, like in Section 2.1 for hierarchical testing. For the ith individual who we assume is located in the jth row and kth column of the array, the complexity is

$$E(C_i) = 1 + P(T_{jk} = 1).$$

The leading 1 in the expression represents the stage one testing of the row and column groups. A stage two test is needed with a probability given by $P(T_{jk} = 1)$.

Kim et al. (2007) also describe a variation of array testing where testing begins by forming one overall group, referred to as a "master pool," in stage one for all individuals represented in the array. If this group tests negative, all individuals represented in it are declared to be negative. Otherwise, array testing is performed as normal for stages two and three. To represent the complexity for array testing with a master pool, define X_0 as the binary testing outcome of the master pool. Also, define T_{2jk} to be the same as T_{jk} with the additional condition of $X_0 = 1$ in each if statement. The complexity for the ith individual is

$$\begin{aligned} E(C_i) &= 1 + P(X_0 = 1) + P(X_0 = 1, T_{2jk} = 1) \\ &= 1 + (1 - S_{p,1})(1-p)^I + S_{e,1}\{1 - (1-p)^I\} + P(X_0 = 1, T_{2jk} = 1) \end{aligned}$$

The expression for $P(X_0 = 1, T_{2jk} = 1)$ was derived by Kim et al. (2007) and Hitt (2020). We also provide this expression in Web Appendix A.

2.3 Sterrett

The Sterrett testing algorithm (Sterrett 1957) was first proposed as a more efficient way to perform group testing than Dorfman testing. Later development by Johnson et al. (1991) took into account testing error. The algorithm continues to be an often cited way to perform group testing with recent papers including Mobini et al. (2023) and Heath et al. (2024). Sterrett testing is hierarchical but takes a quite different approach than the algorithms described in Section 2.1. For this reason, we treat the algorithm separately and describe it next.

The algorithm begins in the same way as Dorfman testing: test portions of all I specimens in the stage one group. If the group tests negative, all I individuals are declared negative. If the group tests positive, the algorithm attempts to use fewer tests to detect the positive individuals than Dorfman testing. This process involves randomly selecting individuals to retest one by one in separate stages until a positive individual is found. Once a positive is found, specimens of the remaining individuals are regrouped and tested. If this group is negative, each corresponding individual is declared as negative. If the group is positive, the process of retesting randomly selected individuals and regrouping continues in the same pattern until all individuals are declared positive or negative. Figure 1 demonstrates this algorithm for $I = 4$. There are at most seven possible stages for this case, and, in general, there can be at most $2I - 1$ stages. While Sterrett testing is generally more efficient than Dorfman testing, this is an example of the need to understand an algorithm's complexity because of the potential for a larger number of stages.

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The ordering of when the ith individual is tested plays an important role in the derivation of the complexity. Corresponding to Figure 1, individual $i = 4$ can be tested only in stages one and two, but a positive or negative outcome for individual $i = 1$ may not be determined until stage seven. For this reason, define $C_i^{(I,j)}$ as the number of stages for the ith individual who is tested in the jth order for a set of I individuals. This number of stages represents the stages until the ith individual is determined to be positive or negative, rather than the number of stages this individual has been included in a test. The expected number of stages in general for the ith individual is $E(C_i) = I^{-1}\sum_{j=1}^{I} E\left(C_i^{(I,j)}\right)$, because each ordering is equally likely. We provide recursive expressions to calculate $E\left(C_i^{(I,j)}\right)$ for $j = 1,\ldots,I-1$ and $I \geq 3$ in Web Appendix A. The derivation takes advantage of the patterns in Sterrett testing by starting with the three specimens case and building up to the I specimens tested. An expression for $j = I$ is not provided because $E(T) = E\left(C_i^{(I,I)}\right)$ due to the duality between the number of tests and the number of stages for the last specimen tested. In this case, the expression for the expected number of tests in Bilder et al. (2010) can be used for the calculation.

3 Evaluation

We examine the efficiency and complexity for the five group testing algorithms at $p = 0.01, 0.02, \ldots, 0.10$. The sensitivity and specificity values are set at 0.99 for each stage, where we discuss other accuracies later. These values for p, sensitivity, and specificity emulate what occurs in group testing practice. Only square arrays ($J = K$) are used for the array-based algorithms because previous research has shown no practical efficiency benefits with non-square arrays (Bilder et al. 2021*b*). The group sizes chosen for each algorithm are those that lead to the best efficiency (smallest $E(T)/I$ for an algorithm). Using these optimized group sizes ensures that the algorithms are compared at their best relative to efficiency, which provides a level of fairness for our comparisons.

The maximum group size is set to 15 for our initial evaluation. The one exception is a master pool used with array testing will have a group size of JK in stage one (J and K are less than or equal to 15). This is because a limit of size 15 for the master pool would lead to an array size to be at most 3×3. Therefore, it is important in testing a master pool to make sure that the dilution effect does not significantly reduce the sensitivity at this stage. If the sensitivity is reduced, our derivations allow for it to be set to a lower value.

Figure 2 provides plots for the efficiency (left side) and complexity (right side). The group sizes used for each algorithm are given in Web Appendix C. With respect to efficiency, each algorithm has an efficiency much less than 1, indicating significant reductions in the expected number of tests needed in comparison to testing each individual separately. Dorfman testing has the worst performance among the algorithms because its $E(T)/I$ is larger than the corresponding values for the others. For example, Dorfman testing has an efficiency of 0.20 at $p = 0.01$. While this represents an 80% reduction in tests on average relative to testing each specimen separately, it is 24% times larger than the efficiency for array testing, which is the

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algorithm with the next closest efficiency value of 0.16. The algorithms with the best efficiency are Sterrett testing at the larger values of *p* and three-stage hierarchical testing at the lower values of *p*. However, the differences among the algorithms can be small, especially at the middle values of *p*.

While Sterrett testing and array testing with master pool are among the most efficient algorithms, they are the most complex of the five as shown in the right-side plot of Figure 2. In particular, Sterrett testing's best-among-algorithms efficiency at the larger values of *p* comes at a cost of high complexity, where an expected number of stages of more than two occurs. This is especially important to note when relating complexity to time. As remarked in Section 1, the CDC's RT-PCR assay for SARS-CoV-2 takes four hours to complete. Therefore, a laboratory may only be able to complete two rounds of testing during a standard eight-hour work day. Sterrett testing is expected to take more than one work day on average then, which delays individuals from receiving their test outcomes. Overall, these results for Sterrett testing and array testing with a master pool should at least make laboratories question whether these algorithms are worthwhile to implement despite their efficiency.

Dorfman testing has the second best complexity, which may be good enough to overlook its efficiency when choosing an algorithm. Overall, array testing has the best complexity. For example, its complexity is 1.03 at $p = 0.01$, which is 7.3% times smaller than the complexity for Dorfman testing and only a little larger than the complexity that occurs when testing each individual separately ($E(C_i) = 1$). The change in complexity from $p = 0.01$ to $p = 0.02$ for array testing with a master pool is because of a large change in the optimal group sizes from 5×5 to 15×15, subject to our maximum group size limit of 15. The actual optimal group sizes are 25×25 for $p = 0.01$ when the limit is removed, and the large change in complexity no longer occurs with this array size.

The plots in Figure 2 also allow one to examine the choice among algorithms at a specific *p* as a constrained minimization problem. Define γ as a constant for the maximum expected number of stages that one can accept and *Q* as the maximum group size possible (excluding the master pool size). The minimization becomes

$$\text{Minimize} E(T)/I \text{ subject to } E(C_i) \le \gamma \text{ and } Q \le 15.$$

For example, three-stage hierarchical testing is best when $p = 0.01$ and $\gamma = 1.5$ stages. While it may be more desirable to have a globally optimal algorithm, Figure 2 and additional plots in Web Appendix C (discussed shortly) show it is not possible. A practical advantage of these plots for laboratories is that it allows them to judge whether differences among algorithms for efficiency and/or complexity are meaningful in practice. Relative to the $p = 0.01$ case, a laboratory may prefer array testing over three-stage hierarchical testing because its complexity is much lower (1.03 vs. 1.19) with a small sacrifice in efficiency (0.16 vs. 0.14).

We provide additional evaluations in Web Appendix C that allow us to examine alternative sensitivity and specificity values. In particular, we find that the same patterns generally occur among algorithms as in Figure 2. Array testing with a master pool can have the best efficiency

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among the five algorithms, but its complexity remains the worst or second worst. Also, complexity tends to decrease as the specificity of the assay increases, with Sterrett testing having the largest decrease. This decrease in complexity occurs because there will be fewer false positive groups as the specificity increases. When fewer groups test positive, fewer stages are needed to complete an algorithm. Trends relative to changing levels of the sensitivity are much more difficult to distinguish because group testing is used in low disease prevalence situations.

In addition to alternative sensitivity and specificity values, we also examine in Web Appendix C the effect that the maximum possible group size has on the efficiency and complexity. Again, our general conclusions about the complexity remain the same.

4 Application

LabCorp first received an EUA for their detection of SARS-CoV-2 via 4×4 and 5×5 arrays in July 2020 (U.S. Food & Drug Administration 2020, LabCorp 2022). In their original EUA summary document (LabCorp 2020), they state array testing was used because Dorfman testing "is not practical to implement in a high throughput testing environment where many thousands of samples would need to be pulled and retested every day." This means that LabCorp believes Dorfman testing is too complex to implement because a second stage would be required much more often than with array testing. It is also important to note that these documents do not provide evidence for their claim or even why their group sizes were chosen. The purpose of our examination here is to determine if there truly is evidence to support the claim using their group sizes and our complexity measure.

With respect to accuracy, the original EUA summary document provides an estimate of 97.7% for a positive percentage agreement with testing specimens separately. While this is not quite the sensitivity of an assay (Fitzpatrick et al. 2021), we will use a first-stage sensitivity of $S_{e,1} = 0.977$ for the application here. We will use $S_{e,2} = 1$ to be consistent with positive percentage agreement and each specimen being tested separately in a second stage. The document also does not provide a specificity value. Therefore, we will use $S_{p,1} = S_{p,2} = 1$ for our comparisons similarly as Abdalhamid et al. (2020). Additional comparisons using different sensitivity and specificity values are discussed shortly.

LabCorp (2020) provides a list of prevalences examined to determine the efficiency of array testing. We include these in Table 1 along with our computed efficiencies and complexities. We also include the same measures for Dorfman testing because of LabCorp's statements about it. Our table shows that both algorithms dramatically reduce the expected number of tests needed in comparison to testing each individual separately, while also having complexities that can be quite close to 1 for lower values of *p*. Our table also shows that array testing has a smaller complexity than Dorfman testing for each *p*. But, array testing can have a much worse efficiency value, by up to two times larger, than Dorfman testing.

Taken alone, our new complexity measure provides formal evidence supporting LabCorp's statement. Thus, while array testing still will have retesting in a second stage, it will occur less often than for Dorfman testing. However, the differences between complexity values at the lower

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values of p are quite small. We expect that most laboratories would prefer Dorfman testing at least for $p = 0.001$ and $p = 0.005$, because half as many tests are needed, on average, with a very small increase in complexity. Therefore, we caution against stating one algorithm is better than another in a numerical manner without accounting for whether their differences are meaningful in practice.

Web Appendix D examines a few additional scenarios for the sensitivity and specificity with group sizes of four and five. The conclusions largely stay the same. The main situation where they differ is for low prevalences and accuracies. In that case, the complexity can be a little larger for array testing than for Dorfman testing. While this behavior is different than what was shown in Section 3, it is important to note that optimal group sizes were used in that section, which is not usually the case here. These results illustrate that a case-by-case evaluation (e.g., at particular values of p) is needed rather than global statements that one algorithm is always better than another.

5 Discussion

Our proposed complexity measure provides a new tool for deciding which group testing algorithm to put into practice. When examined together, the efficiency and the complexity allow laboratories to understand how best to save testing resources and increase testing capacity, while accounting for aspects of implementation. Our supporting materials provide R functions to calculate both measures. Because individuals at laboratories without statistical training may want to perform these calculations, we have also added complexity calculations to our "A Shiny app for Pooled Testing" (chrisbilder.com/app) that was widely used during the COVID-19 pandemic to calculate efficiency. Furthermore, the complexity can become a fundamental aspect of group testing research when assessing newly proposed algorithms. In particular, this measure can help researchers understand if their proposal, which could look great on paper, may actually not be practical for laboratories.

Of the five group testing algorithms examined in our work, there was not one that was best for both efficiency and complexity. Therefore, we recommend calculating both measures for the algorithms of interest corresponding to the diagnostic testing situation. In the end, laboratories may face a trade off between efficiency and complexity when choosing between algorithms. Along with our recommendation, there are two specific overall results that are important to focus on. First, Dorfman testing is the most widely used algorithm by laboratories despite other algorithms frequently having better efficiency. Our research shows that laboratories can now point to its lower complexity as a potential reason for giving up some efficiency gains. Second, array testing is much less widely used than Dorfman testing in practice, despite its efficiency benefits being known to statistics researchers for some time (Phatarfod and Sudbury 1994; Kim et al. 2007). Its low complexity shown in our paper provides laboratories with new evidence to support its wider adoption.

With respect to the statement from Fenichel et al. (2021) that was included in Section 1, we expect there are additional issues beyond the number of stages that they are referencing. Many of these issues cannot be measured in a statistical manner. For instance, the group formation process needs to be considered. Laboratories with modern robotic pipetting instruments will have

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no problems forming groups for the algorithms discussed in this paper. However, some laboratories perform pipetting by hand when forming groups. Complicated non-hierarchical testing algorithms (those beyond array testing) just may not be possible in these situations. Furthermore, groups are at times formed outside the laboratory while at a specimen collection site (e.g., Ciaranello et al. 2021 and https://covidedtesting.com), which can make algorithms like Dorfman testing more desirable. Another issue involves laboratory information systems that track the positive/negative test results. Some systems are not designed for group testing, and we have seen group test results tracked by hand in this situation. Lastly, space at times can become an issue when storing specimens for retesting, depending on laboratory resources and testing loads. Overall, our complexity measure provides a new tool for laboratories to use that can help them better understand group testing algorithms. This measure can also help group testing researchers who are outside of a laboratory to understand better the complexities of implementation.

Supplementary materials

Additional materials may be found in the online version of the article at the publisher's website and at chrisbilder.com/grouptesting/complexity. These materials include our web appendices and R programs.

Disclosure statement

The authors report that there are no competing interests to declare.

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Data availability

The sensitivity data are available from LabCorp (2022).

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Figure 1: Sterrett algorithm for $I = 4$. The four individuals are denoted as 1, 2, 3, and 4. While the algorithm randomly selects one individual when a group tests positive, we display the algorithm by selecting the individuals in reverse numerical order. Braces are used to denote individuals being tested as a group. "None" indicates no further tests are needed.

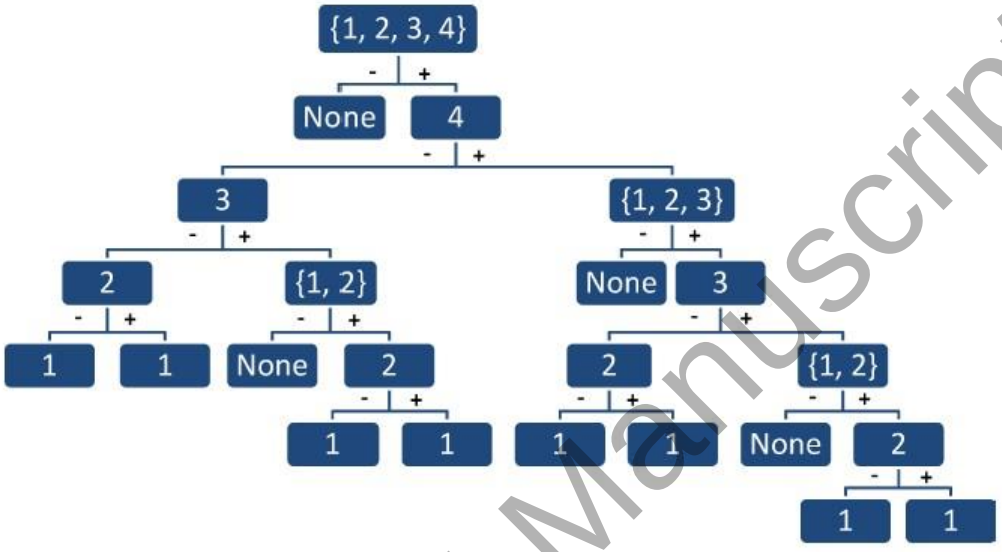

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Figure 2: Efficiency and complexity when $S_{e,d} = 0.99$ and $S_{p,d} = 0.99$ at each stage *d*.

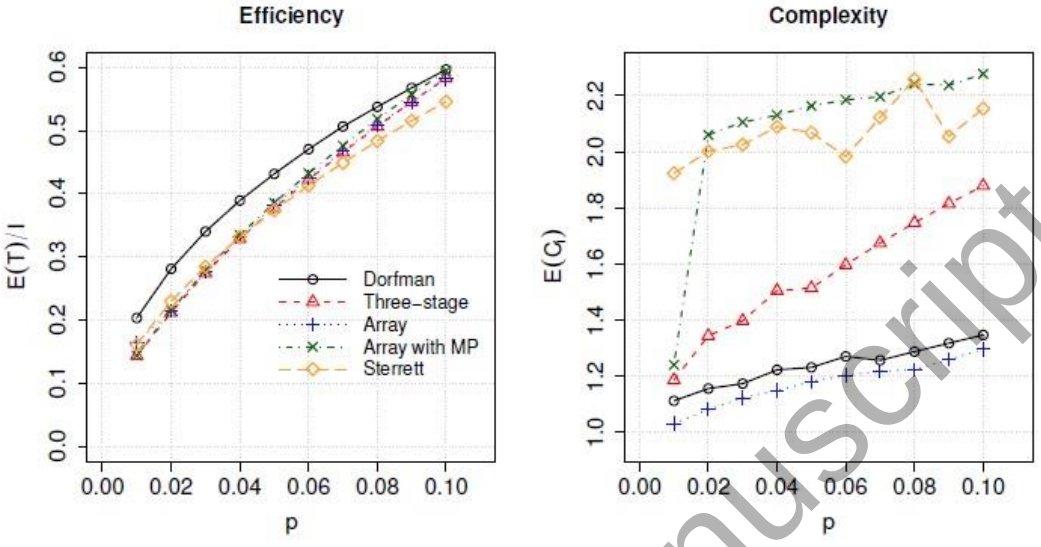

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Table 1: Efficiency and complexity for the LabCorp implementation of group testing. Shaded cells are for the algorithm that has the lower value with respect to its efficiency or complexity.

	Dorfman ($I = 4$)		Array ($J = K = 4$)	
p	Efficiency	Complexity	Efficiency	Complexity
0.001	0.25	1.00	0.50	1.00
0.005	0.27	1.02	0.51	1.01
0.01	0.29	1.04	0.51	1.01
0.03	0.36	1.11	0.54	1.04
0.05	0.43	1.18	0.57	1.07
0.10	0.58	1.33	0.66	1.16
0.15	0.71	1.46	0.77	1.27

	Dorfman ($I = 5$)		Array ($J = K = 5$)	
p	Efficiency	Complexity	Efficiency	Complexity
0.001	0.20	1.00	0.40	1.00
0.005	0.22	1.02	0.41	1.01
0.01	0.25	1.05	0.41	1.01
0.03	0.34	1.14	0.44	1.04
0.05	0.42	1.22	0.48	1.08
0.10	0.60	1.40	0.60	1.20
0.15	0.74	1.54	0.73	1.33

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